

CUSTOMER CHURN ANALYSIS & INTELLIGENCE

End-to-End Data Analytics & Business Intelligence Project

SQL • Python • Pandas • NumPy • Matplotlib • Seaborn • Power BI • DAX

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Project Overview

An end-to-end customer churn analytics project focused on identifying customer retention patterns, analyzing subscription and contract behavior, evaluating customer-support signals, and translating analytical findings into actionable business recommendations.

The project combines **SQL, Python-based exploratory data analysis, customer segmentation, churn-risk analysis, and an interactive Power BI dashboard** to provide a complete view of customer churn and retention.

Business Objective

Identify who is churning, understand the patterns associated with churn, determine when churn is occurring, quantify customer/revenue exposure, and recommend data-driven retention actions.

Data Analytics Portfolio Project | 2026

BUSINESS PROBLEM & OBJECTIVES

1. Business Problem

Customer churn directly impacts recurring revenue, customer lifetime value, and long-term business growth.

The objective of this project is to analyze customer, subscription, support, satisfaction, and cancellation data to identify the customer segments showing higher observed churn and understand the factors associated with customer attrition.

The analysis focuses on three fundamental questions:

WHO?

Identify which customers and customer segments have churned or show higher churn risk.

WHY?

Analyze customer behavior, subscription characteristics, support interactions, satisfaction, and cancellation reasons associated with churn.

WHEN?

Identify periods or customer lifecycle patterns where churn is more concentrated.

Business Questions

The analysis was designed to answer the following questions:

1. What is the overall customer churn rate?
 2. How many customers have churned versus been retained?
 3. Which subscription plans have the highest observed churn?
 4. How does churn differ between monthly and annual contracts?
 5. Do churned customers have different satisfaction levels?
 6. Are support escalations associated with churn?
 7. What are the major observed cancellation reasons?
 8. Which geographic segments show higher observed churn?
 9. When does churn appear to be concentrated?
 10. What revenue and customer-value exposure is associated with churn?
 11. Which customers should be prioritized for retention?
 12. What actions can the business take to improve retention?
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DATASET & DATA MODEL

2. Dataset & Data Model

The project uses customer, subscription, and support-related information to create an analytical view of customer churn.

The data is organized into three logical areas:

Customer Data

Contains customer demographic and geographic information.

Key fields:

- Customer ID
 - Customer Name
 - Gender
 - Date of Birth
 - Country
 - State
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Subscription Data

Contains customer subscription and financial information.

Key fields:

- Customer ID
- Subscription Start Date
- Subscription Type
- Plan Type
- Contract Type
- Renewal Date
- Cancellation Date
- Cancellation Reason

- Monthly Charges
 - Customer Lifetime Value
 - Churn Score
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Support Data

Contains customer service and satisfaction information.

Key fields:

- Customer ID
 - Complaint Date
 - Complaint Count
 - Escalations
 - CSAT Score
 - Support Comments
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DATA CLEANING & QUALITY CHECKS

3. Data Preparation

Before performing analysis, the dataset was reviewed and prepared for analytical use.

Data Cleaning Activities

1. Column Standardization

Column names were reviewed and standardized to make them easier to query and use in Python and Power BI.

2. Data Type Validation

Date, numeric, categorical, and indicator fields were checked to ensure that they were stored using appropriate data types.

3. Missing Value Checks

The dataset was checked for missing or null values that could affect KPI calculations and analysis.

4. Duplicate Checks

Customer records were reviewed to ensure that duplicate customer records did not distort customer-level KPIs.

5. Categorical Validation

Important categorical fields were reviewed, including:

- Plan Type
- Contract Type
- Subscription Type
- Gender
- Cancellation Reason
- Churn Status

6. Boolean/Indicator Validation

Fields such as churn status and escalation indicators were validated before analysis.

Quality Control

The following quality checks were performed before creating the dashboard:

- Record count validation
- Customer ID uniqueness check
- Null-value checks
- Data-type checks
- Date validation
- Category consistency checks
- Churn-flag validation
- Revenue/KPI calculation validation

FEATURE ENGINEERING & KPI FRAMEWORK

4. Feature Engineering & KPI Development

Feature engineering was performed to create analytical variables and business KPIs required for churn analysis.

Churn Flag

A churn indicator was used to separate customers into:

Churned

Customers with a recorded cancellation.

Retained

Customers without a recorded cancellation.

Tenure

Customer tenure represents the length of the customer's subscription relationship.

It is calculated using subscription start date and the relevant observation/cancellation date.

Core KPIs

Total Customers

Number of unique customers in the dataset.

Churned Customers

Number of unique customers classified as churned.

Retained Customers

Number of unique customers classified as retained.

Churn Rate

Churn Rate = Churned Customers / Total Customers × 100

Retention Rate

Retention Rate = 100 – Churn Rate

ARPU

Average revenue per relevant customer based on the project's revenue definition.

Revenue Exposure

Revenue associated with churned customers based on monthly charges/revenue fields used in the analysis.

Customer Lifetime Value

Estimated value associated with the customer relationship using the project's CLTV calculation.

Escalation Rate

Percentage of customers with support escalation activity.

CSAT

Customer satisfaction score used to compare customer experience between retained and churned customers.

PYTHON EDA

5. Exploratory Data Analysis

Python was used to explore customer behavior and identify patterns before building the Power BI dashboard.

Tools Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
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EDA Techniques

The analysis included:

Aggregation

Used to calculate customer counts, revenue, churn, and other KPIs.

Group By Analysis

Used to compare churn across:

- Plan Type
- Contract Type
- State
- Churn Status
- Customer segments

Pivot Analysis

Used to compare multiple customer dimensions and identify patterns.

Distribution Analysis

Used to understand customer-value and satisfaction-related variables.

Time-Based Analysis

Used to identify periods with higher observed cancellation activity.

Key EDA Findings

The exploratory analysis identified several important patterns:

- Basic-plan customers have the highest observed churn rate.
- Monthly-contract customers show substantially higher churn than annual-contract customers.
- Churned customers have lower average CSAT.
- Cancellation reasons include competitor switching, pricing/value concerns, content availability, trial cancellation, and service quality.
- September 2024 recorded the highest observed cancellation activity.

- Karnataka showed the highest observed state-level churn rate in the current dataset.
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POWER BI DASHBOARD

6. Power BI Dashboard

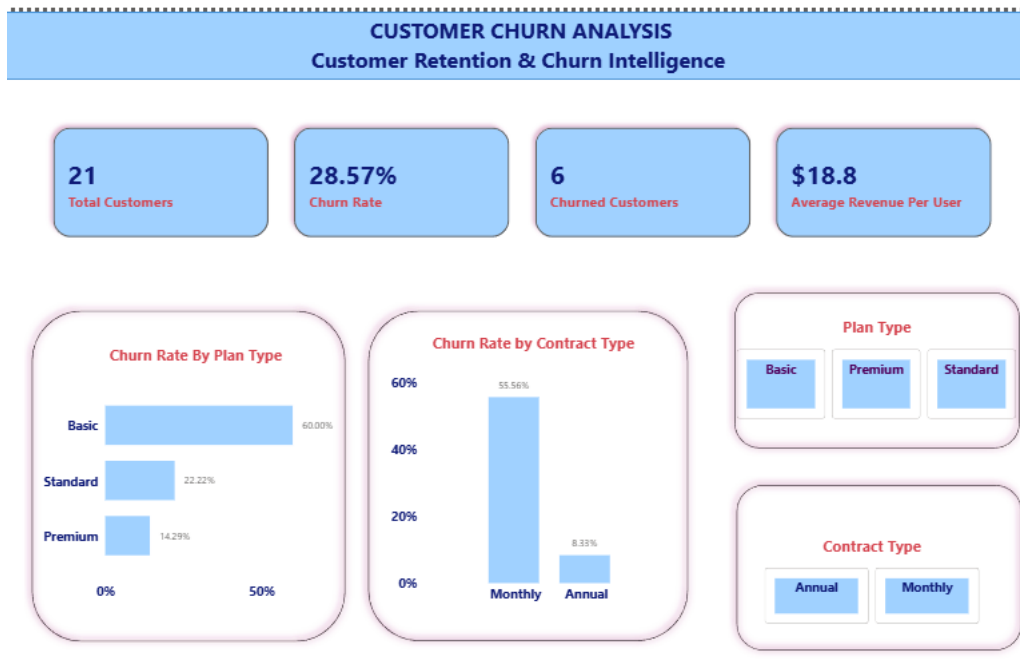
An interactive Power BI dashboard was developed to convert the analytical results into a business-friendly reporting interface.

Dashboard Objectives

The dashboard allows users to:

- Monitor overall churn
 - Compare retained and churned customers
 - Analyze churn by plan
 - Analyze churn by contract
 - Monitor customer satisfaction
 - Analyze support escalations
 - Review cancellation reasons
 - Explore geographic churn patterns
 - Monitor customer-value indicators
 - Identify higher-risk customer segments
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Dashboard 1 — Executive Churn Overview



Purpose

Provides an executive-level overview of customer volume, churn, retention, plan-level churn, contract-level churn, and customer/revenue indicators.

Key KPIs

- Total Customers
- Churned Customers
- Retained Customers
- Churn Rate
- Retention Rate
- Revenue
- CLTV
- Monthly Charges

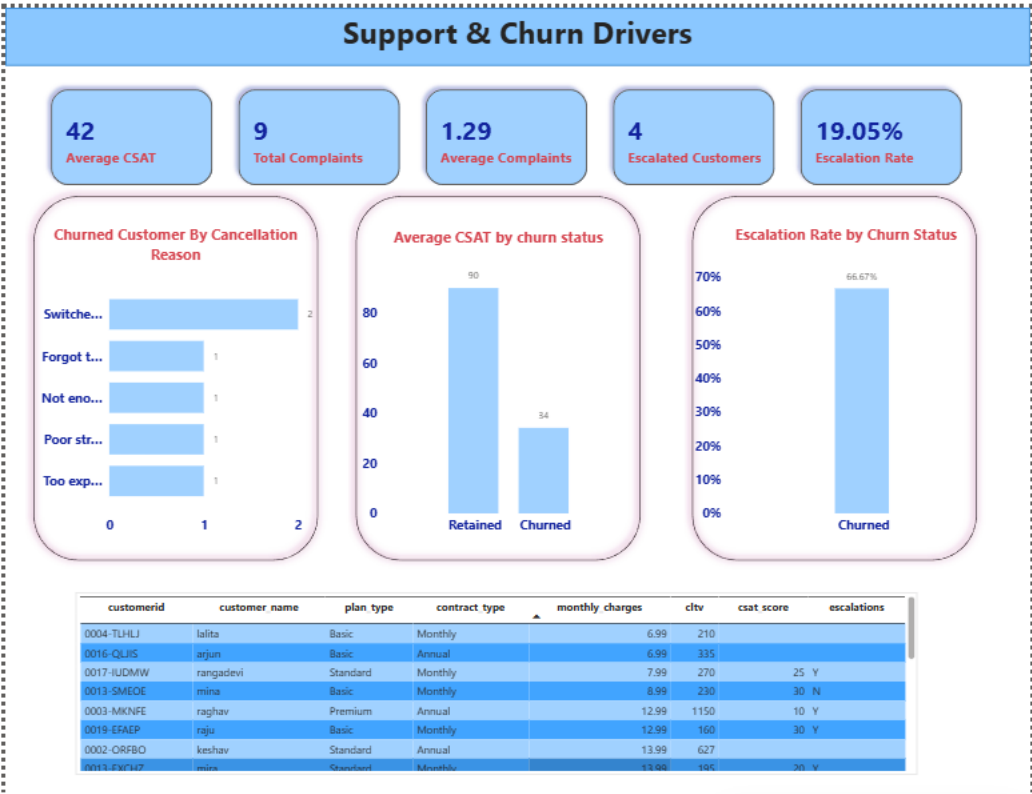
Dashboard 2 — Customer & Support Intelligence



Purpose

Analyzes churn through customer satisfaction, complaints, support escalations, cancellation reasons, and customer segments.

Dashboard 3 — Churn Analysis / Customer Segmentation



Purpose

Provides deeper analysis of churn patterns across customer segments and risk indicators.

KEY BUSINESS INSIGHTS

7. Key Business Insights

Insight 1 — Overall Churn

The dataset contains **21 customers**, of which **6 customers have churned**.

Observed Churn Rate

28.57%

Retention Rate

71.4%

Business Interpretation

The dataset indicates a meaningful customer-retention challenge and provides a baseline for identifying higher-risk customer segments.

Important: This is a dataset-level observation and should not be interpreted as the churn rate of a real company's entire customer population.

Insight 2 — Basic Plan Has the Highest Churn

Plan	Observed Churn Rate
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Basic	60.00%
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Standard	22.22%
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Premium	14.29%
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Finding

Basic-plan customers have the highest observed churn rate at **60%**.

Recommendation

Investigate:

- Pricing/value perception
- Feature availability
- Content offering
- Service quality
- Competitor positioning
- Upgrade opportunities

The analysis does not establish that price alone caused the churn.

Insight 3 — Monthly Contracts Have Higher Churn

Contract Observed Churn Rate

Monthly **55.56%**

Annual **8.33%**

Finding

Monthly-contract customers show a substantially higher observed churn rate than annual-contract customers.

Business Recommendation

The business could test:

- Annual-plan incentives
 - Loyalty benefits
 - Targeted retention campaigns
 - Renewal reminders
 - Personalized upgrade/migration offers
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CUSTOMER EXPERIENCE & CANCELLATION INSIGHTS

8. Customer Satisfaction & Support Analysis

CSAT and Churn

Average CSAT in the dashboard indicates:

Customer Status Average CSAT

Retained ≈42

Churned ≈34

Finding

Churned customers show lower average satisfaction than retained customers.

Business Interpretation

Lower customer satisfaction appears to be associated with churn in this dataset and can be treated as a potential early-warning signal.

Recommendation

Prioritize customers with:

- Low CSAT
- Recent complaints
- Support escalations
- High churn score

for proactive customer-success follow-up.

Support Escalations

The dashboard reports an overall escalation rate of approximately:

19.05%

Finding

Escalation activity appears in the churn-status analysis and should be monitored alongside CSAT and complaints.

Recommendation

Create an escalation follow-up process so that customers experiencing unresolved issues receive proactive support.

Analytical caution: The dataset indicates an association worth investigating. It does not prove that support escalation causes churn.

Cancellation Reasons

Observed cancellation reasons include:

- Switched to competitor
- Forgot to cancel trial
- Not enough content
- Poor streaming quality
- Too expensive

Business Interpretation

Customer cancellations appear to involve multiple potential drivers, including:

Competition + Pricing/Value + Content + Service Quality

Recommendation

Use cancellation reasons to create targeted retention strategies instead of applying the same offer to every customer.

CHURN TIMING, GEOGRAPHY & REVENUE IMPACT

9. Churn Timing

September 2024

The dataset shows the highest observed cancellation activity in **September 2024**.

Recommended Investigation

Investigate whether any of the following occurred around this period:

- Pricing changes
- Product changes
- Content changes

- Technical/service issues
- Promotional changes
- Competitor activity
- Renewal-related events

The dashboard identifies September as a period requiring investigation; it does not prove what caused the increase.

Geographic Churn

Karnataka recorded the highest observed state-level churn rate in the current dataset.

Recommended Investigation

Investigate:

- Customer complaints
- Technical issues
- Pricing changes
- Service experience
- Regional operational issues
- Competitor activity

Important Limitation

Because the dataset contains only **21 customers**, state-level percentages can be highly unstable.

For example, if a state has only one customer and that customer churns, the observed churn rate would be 100%.

Therefore:

Geographic findings should be treated as exploratory signals rather than reliable population-level conclusions.

Revenue & Customer Value Impact

Total Revenue

₹395

Revenue Associated with Churned Customers

₹74

Estimated CLTV Associated with Churned Customers

₹2,047

Revenue Exposure

₹74 / ₹395 ≈ 18.7%

Interpretation

The churned cohort represents a measurable portion of the revenue/value captured in this dataset.

These values should be interpreted according to the project's revenue and CLTV calculation definitions.

RISK PRIORITIZATION & ACTION PLAN

10. Customer Churn Risk Prioritization

Customer retention should not be based on churn probability alone.

A more useful business approach is to combine:

Churn Risk + Customer Value + Customer Experience

Priority Customer Signals

High Churn Score

Customers showing higher predicted/assigned churn risk.

High CLTV

Customers representing greater potential long-term value.

Low CSAT

Customers showing weaker satisfaction.

Complaints

Customers with recent or repeated complaints.

Escalations

Customers requiring additional support intervention.

Monthly Contract

Customers belonging to the highest observed contract-level churn segment.

Recommended Retention Priority

High Churn Risk

+

High Customer Value

+

Low CSAT / Complaints / Escalations

↓

HIGH PRIORITY RETENTION

Recommended Actions

Priority 1: Proactively contact high-risk, high-value customers.

Priority 2: Address unresolved complaints and escalations.

Priority 3: Target monthly customers with annual-plan incentives.

Priority 4: Investigate Basic-plan value and retention issues.

Priority 5: Use cancellation reasons to personalize retention offers.

FINAL RECOMMENDATIONS

11. Business Recommendations

Priority Finding		Recommended Action
1	Monthly churn = 55.56%	Develop annual-plan migration and retention campaigns
2	Basic churn = 60%	Investigate pricing, features, content and value
3	Churned CSAT ≈ 34 vs retained ≈ 42	Proactively engage low-CSAT customers
4	Escalation rate = 19.05%	Create faster escalation-resolution workflows
5	Competitor switching observed	Monitor competitor pricing, features and offers
6	September 2024 has highest cancellations	Investigate events/changes during this period
7	Karnataka has highest observed state churn	Investigate regional issues, while validating with more data
8	High/Medium churn-risk customers	Prioritize based on risk + CLTV + customer experience

Strategic Retention Framework

IDENTIFY

Find customers with higher churn risk.



PRIORITIZE

Combine churn risk with CLTV and customer-experience signals.



INTERVENE

Use targeted retention actions.



MONITOR

Track churn, retention, CSAT and revenue impact.



OPTIMIZE

Improve retention strategies based on measured outcomes.

PROJECT LIMITATIONS & NEXT STEPS

12. Analytical Limitations

This project is based on a relatively small dataset containing **21 customer records**.

Therefore:

- Churn percentages should be treated as dataset-level observations.
 - Small customer segments can produce extreme percentages.
 - Geographic comparisons require a larger sample.
 - Observed relationships do not establish causation.
 - Revenue and CLTV figures depend on the project's calculation definitions.
 - Churn-risk scoring should be validated against a larger historical dataset before being used operationally.
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Future Improvements

With a larger dataset, the analysis could be extended to include:

1. Historical Churn Analysis

Analyze customer behavior over multiple months or years.

2. Cohort Analysis

Compare retention across acquisition and subscription cohorts.

3. Customer Lifetime Analysis

Analyze CLTV by plan, contract and customer segment.

4. Churn Prediction

Develop a machine-learning model to predict future churn.

5. A/B Testing

Measure the effectiveness of retention campaigns.

6. Automated Dashboard Refresh

Connect Power BI directly to the database for recurring reporting.

7. Customer-Level Retention Queue

Create a prioritized list of customers requiring intervention.

PROJECT SUMMARY / GITHUB FINAL PAGE

13. Project Summary

This project demonstrates an end-to-end Data Analytics workflow:

Raw Customer Data



SQL Database



Data Extraction



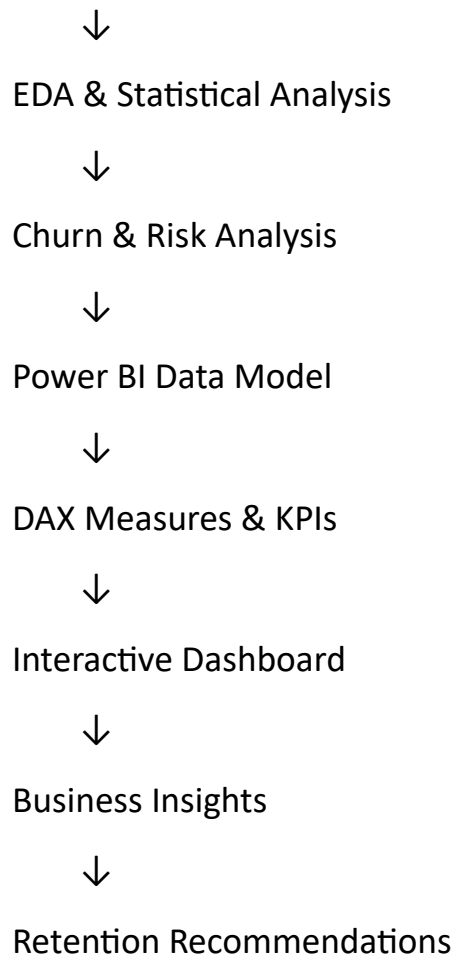
Python / Pandas



Data Cleaning & Quality Checks



Feature Engineering



Technical Skills Demonstrated

SQL

- Data extraction
- Joins
- Aggregations
- GROUP BY
- Filtering
- KPI calculations
- Customer-level analysis

Python

- Pandas

- NumPy
- Data cleaning
- Feature engineering
- Exploratory Data Analysis
- Aggregation
- Visualization

Visualization

- Matplotlib
- Seaborn
- Power BI
- DAX
- KPI cards
- Interactive filters
- Dashboard navigation

Business Analytics

- Churn analysis
- Customer segmentation
- Risk prioritization
- Customer experience analysis
- Revenue exposure analysis
- CLTV analysis
- Retention strategy

Final Project Outcome

The project transformed raw customer, subscription, and support data into an interactive churn intelligence solution. The analysis identified high-risk customer segments, quantified observed churn and customer-value exposure,

highlighted differences between monthly and annual contracts, and translated the findings into targeted retention recommendations.

Author

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Aspiring Data Analyst

Skills: SQL | Python | Power BI | Excel | Pandas | NumPy | Data Visualization | EDA | Business Analytics